

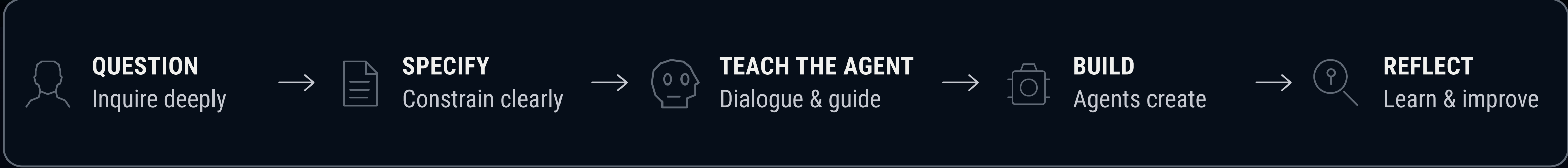


PROJECT MAIEUTIC / 學問

From Inquiry to Software

The CityUHK Initiative on
Specification-Driven & Agentic Computing Education

“The art of bringing
latent knowledge
into light through
questioning.”



FROM FIRST YEAR ONWARD,
STUDENTS LEARN TO
THINK, SPECIFY, AND
BUILD WITH AI AGENTS.

EPMHΣ · AGENT CLIENT PROTOCOL · ASYNC PYTHON

A messenger for your agents

`pip install hermes-acp-sdk`

An async Python client for the Agent Client Protocol. Two context managers stand in for the whole stdio handshake — spawn the agent, open a session, and read a stream of *typed events* while your application never once holds an API key.

I THE OLD ROAD

raw ACP by hand

```
→ {"jsonrpc": "2.0", "id": 1, "method": "initialize", ...}
← {"result": {"protocolVersion": 1, "authMethods": [...]}}
→ {"method": "session/new", "params": {"cwd": "..."}}
→ {"method": "session/prompt", "params": {...}}
← {"update": {"sessionUpdate": "agent_thought_chunk", ...}}
← {"update": {"sessionUpdate": "agent_message_chunk", ...}}
```

spawn the subprocess

track request ids

guess update shapes

negotiate auth methods

reassemble chunks

tear the process down

≈200

LINES OF PLUMBING
BEFORE THE FIRST TOKEN

III THE MESSAGE ARRIVES TYPED

II THE NEW ROAD

hermes-acp-sdk

```
async with HermesClient() as hermes:
    async with hermes.session() as s:
        async for ev in s.prompt(q):
            handle(ev)
```

HermesClient() spawns the agent and shakes hands

session() chooses the model and the profile

prompt() yields typed events as they arrive

3

LINES TO THE SAME
FIRST TOKEN

Fig. 1 — one prompt, end to end

events = []

`async with HermesClient() as hermes:
 print(f"connected to: {hermes.agent_name} {hermes.agent_version}\n")
 async with hermes.session() as s:
 async for ev in s.prompt("Think it through, then answer in one line: what is a Python traceback?"):
 events.append(ev)

print("events received :", dict(Counter(type(e).__name__ for e in events)))
thoughts = "".join(e.text for e in events if isinstance(e, AgentThought))
answer = "".join(e.text for e in events if isinstance(e, AgentText))
usage = [e for e in events if isinstance(e, Usage)]
if thoughts:
 print("\n🧠 reasoning >>>", thoughts[:180].strip(), "...")
print("\n💬 answer >>>", answer.strip())
print("\n📊 usage >>>", usage[-1] if usage else "n/a")`

connected to: hermes-agent 0.18.2

events received : {'Usage': 3, 'AgentThought': 51, 'AgentText': 22, 'Finished': 1}

🧠 reasoning >>> The user asks: "Think it through, then answer in one line: what is a Python traceback?"

This is a simple conceptual question. I should think through it then answer in one line. Le ...

💬 answer >>> A Python traceback is the stack report Python prints when an exception is raised, listing the chain of function calls from the entry point down to the exact line that triggered the error so you can trace and fix the failure.

📊 usage >>> Usage(input_tokens=9863, output_tokens=157, total_tokens=10020, used=None, size=None)

36

MODELS OFFERED
TO ONE SESSION

0

API KEYS HELD
BY YOUR APP

2

CONTEXT MANAGERS,
THE WHOLE API

“Simplicity is a prerequisite for
reliability.”

EDSGER W. DIJKSTRA

IV EVERY APP ITS OWN SANCTUARY

Fig. 2 — one flag per guarantee

```
async with HermesClient(profile="paper-demo", clone_provider=True, auto_prefix=True) as hermes:
    async with hermes.session() as s:
        print("profile : app-paper-demo (its own memory, sessions, skills)")
        print("provider : cloned from the host – the app never handled an API key")
        print("models offered :", len(s.available_models))
        out = []
        async for ev in s.prompt("Reply with exactly: ISOLATED PROFILE OK"):
            if isinstance(ev, AgentText): out.append(ev.text)
        print("agent reply :", "".join(out).strip())

profile : app-paper-demo (its own memory, sessions, skills)
provider : cloned from the host – the app never handled an API key
models offered : 36
agent reply : ISOLATED PROFILE OK
```

V THE LEXICON

HermesClient	spawn · handshake
session()	model · memory
prompt()	async iterator
AgentThought	reasoning chunks
AgentText	the answer
Usage	input · output tokens
Finished	turn complete

Each application receives its own profile — separate memory, sessions and skills — while the provider is cloned from the host, so no credential ever reaches application code.

```
profile="paper-demo"
clone_provider=True
auto_prefix=True
s.available_models
```

BUILT BY AN UNDERGRADUATE

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[GITHUB.COM/VOIXKZ/HERMES-ACP-SDK](#)

REFERENCES

[1] Zhao, C., Wang, Z., Ahmedani, A. A., **Karimtayev, B.**, Marcello-Raphael-Christiano, Lua, N. M. B., Khamza, R. & Chan, C. (2026). DIVE into agentic coding: Can you teach AI to teach you? In *Proceedings of the IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)*. Accepted.

[2] **Karimtayev, B.** (2026). hermes-acp-sdk [Computer software]. PyPI. <https://pypi.org/project/hermes-acp-sdk/>

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